Seminar Presentation
on

Title: A supplier selection decision model using multi-criteria decision analysis in a small manufacturing company

Paper: Rodrigues, Šírová & Mugurusi (2022), IFAC-PapersOnLine 55(10), pp. 2773-2778

by

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INTRODUCTION

- A small Czech company (about **400 workers**) makes parts for the car industry.
- **Copper rod** is its most critical spare part (found by ABC analysis). No rod in stock means the **production line stops**.
- Problem: spare parts are costly and suppliers are often late.
- **Three suppliers** can sell the rod, and **three managers** must agree on one.
- Until now the supplier was chosen by experience only.
- **Aim of the paper:** a clear scoring model to choose the supplier, solved with **AHP** and **ANP**.

Supplier	Located in	What the paper says
Supplier 1	Germany	Best quality, highest price, farthest away
Supplier 2	Czech Republic (Central Bohemia)	Low price, fastest delivery
Supplier 3	Czech Republic (North Bohemia)	Low price, weakest quality

Maintenance manager (4 yrs)
Leads the purchase, negotiates price and delivery

Production manager (5 yrs)
A line stop costs more than any delivery charge

Quality manager (10 yrs)
Zero-defect policy: quality above price and time

Price **€230 to €350** per rod | Delivery **3 to 7 days** | Bad parts **6.67% to 15%** (2018) | Spend about **€25,300** a year

Source: Rodrigues, Šírová & Mugurusi (2022), Sections 3 and 4.

WHAT THE PAPER DID

AHP (Analytic Hierarchy Process)

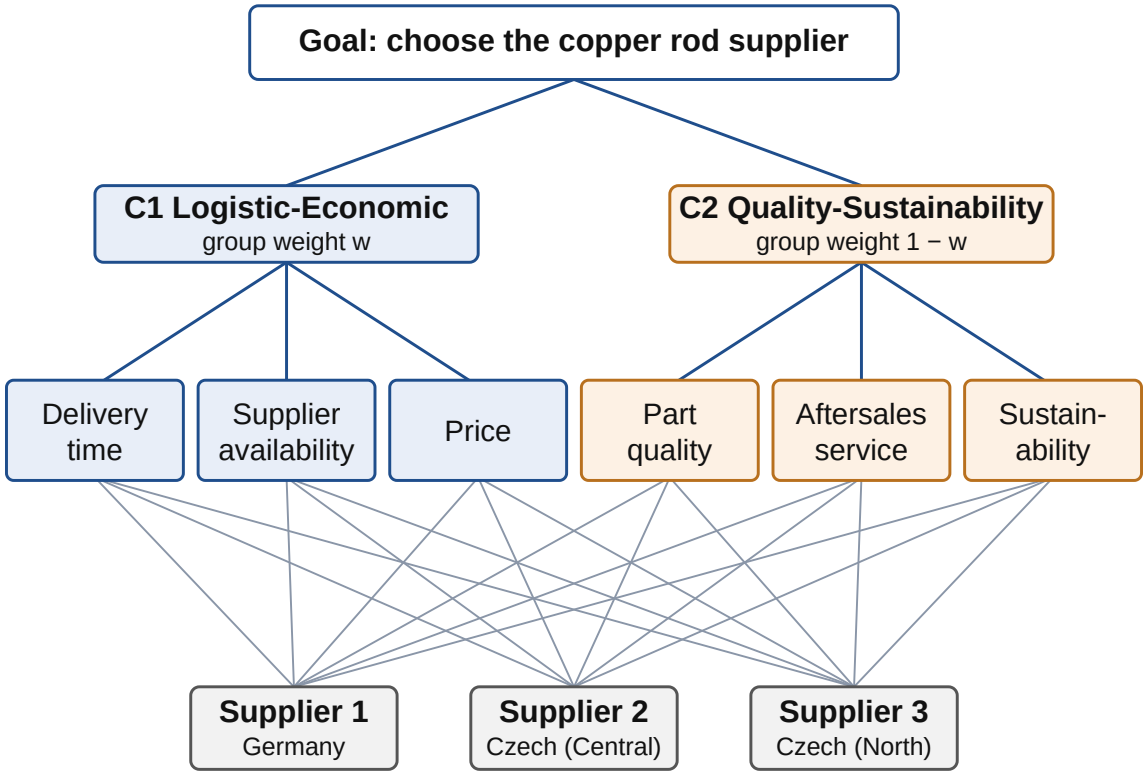


Fig. 1: AHP model. A tree: the two groups are equally important (w = 0.5).

ANP (Analytic Network Process)

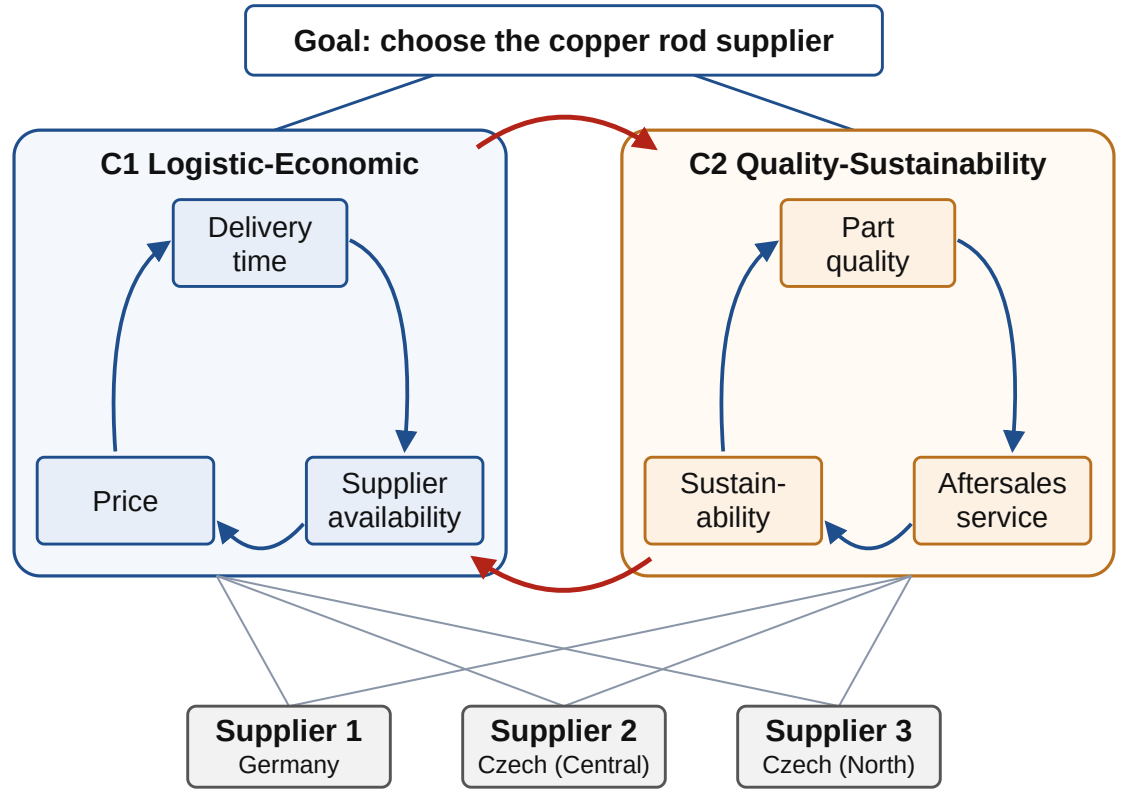


Fig. 2: ANP model. A network: the two groups affect each other (red arrows), and so do the attributes inside each group (blue arrows).

Result: AHP picks **Supplier 2**, ANP picks **Supplier 1**. The paper recommends the ANP answer.

Source: redrawn from Rodrigues et al. (2022), Figures 1 and 2; results from Tables 10 and 11.

GAPS IDENTIFIED IN THE PAPER

Gap	What I found from the paper's own tables
1. Two methods, two winners	I rebuilt both results exactly. **ANP equals plain AHP** with the quality group made **5 times** more important (w = 1/6).
2. No sensitivity check	The winner changes at **w = 0.442**, very close to the w = 0.5 used by AHP (Fig. 3).
3. Answers not consistent	**4 of 8 tables** are above Saaty's 5% limit (8.25%, 5.16%, 6.24%, 5.16%).
4. Measured values given as opinions	Supplier 1 costs only **1.52 times** more (€350 vs €230), but its price score is **7.3 times** lower.
5. One fixed ranking	No check of how sure the winner is when answers vary a little.
6. No cost view	The ranking ignores money lost on bad rods and line stops.

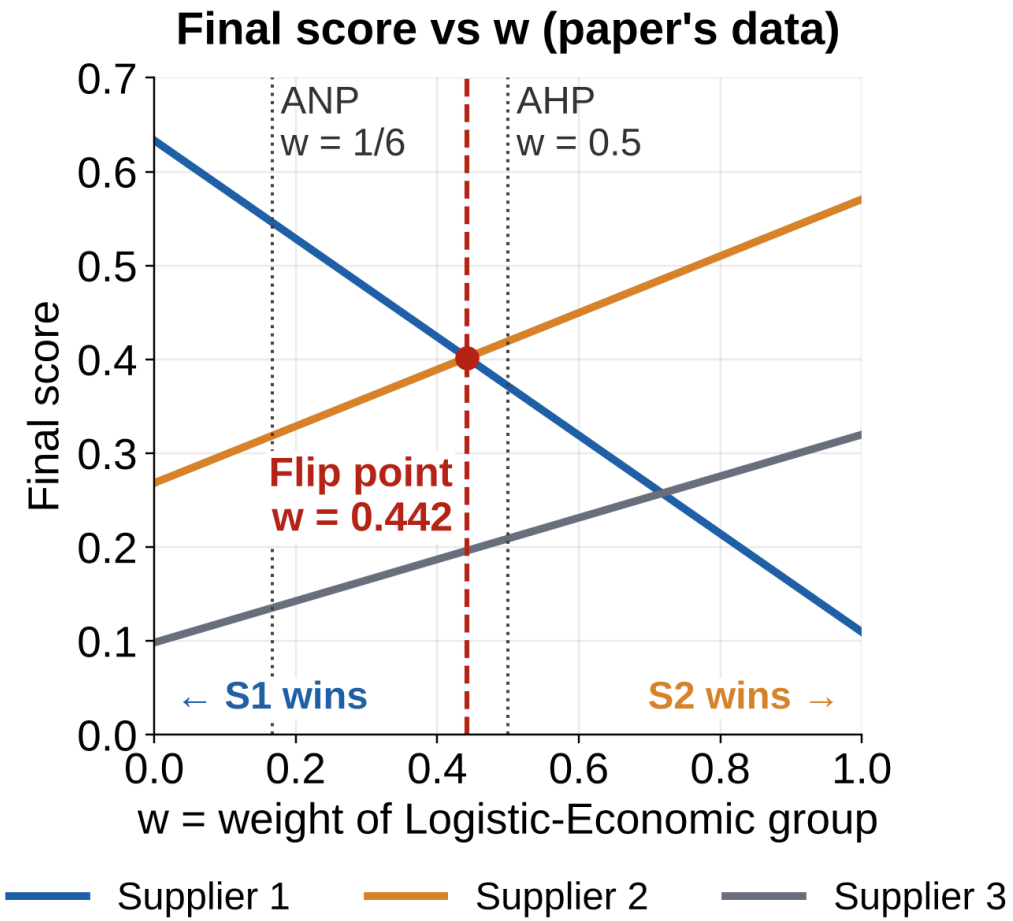


Fig. 3: Same tables, only w changes. Below 0.442 Supplier 1 wins, above it Supplier 2.

Source: own recalculation from Rodrigues et al. (2022), Tables 2 to 12.

MY PROPOSED IMPROVEMENT

Same input values as the paper. Only the way they are used is improved.

1. **Re-check doubtful answers** (gap 3)
   In each failing table, move one answer by one step, so all 8 tables pass.

2. **Use real numbers where they exist** (gap 4)
   Price, delivery days and defect rate: score = (1/x) ÷ Σ(1/x), which keeps the true ratio.

3. **Show the flip point** (gaps 1, 2)
   Managers see exactly which group weight changes the winner.

4. **Test how sure the winner is** (gap 5)
   5,000 trials: each answer moved by ±1 step, w picked at random from 0.1 to 0.9.

5. **Add a 2-year cost per buying plan** (gap 6)
   Only S1, only S2, only S3, and a 50/50 split of S1 and S2.

Answers to re-check

Table	Answer	Change	CR after
T2	Price vs Availability	1/3 → 1/2	**2.36%**
T3	Aftersales vs Sustain.	2 → 1	**0.00%**
T5	Quality: S1 vs S2	3 → 2	**1.36%**
T7	Aftersales: S2 vs S3	2 → 1	**0.00%**

2-year cost of a plan

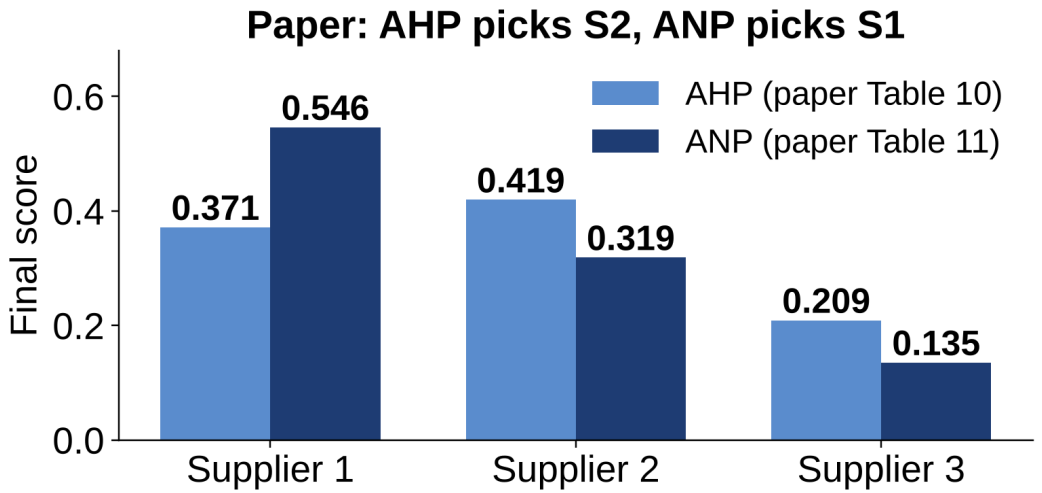
Cost = rods × price
+ bad rods × €150
+ line-stop hours × €300

Price, delivery and defect values from the paper's ranges; €150 and €300 are estimates the company can replace.

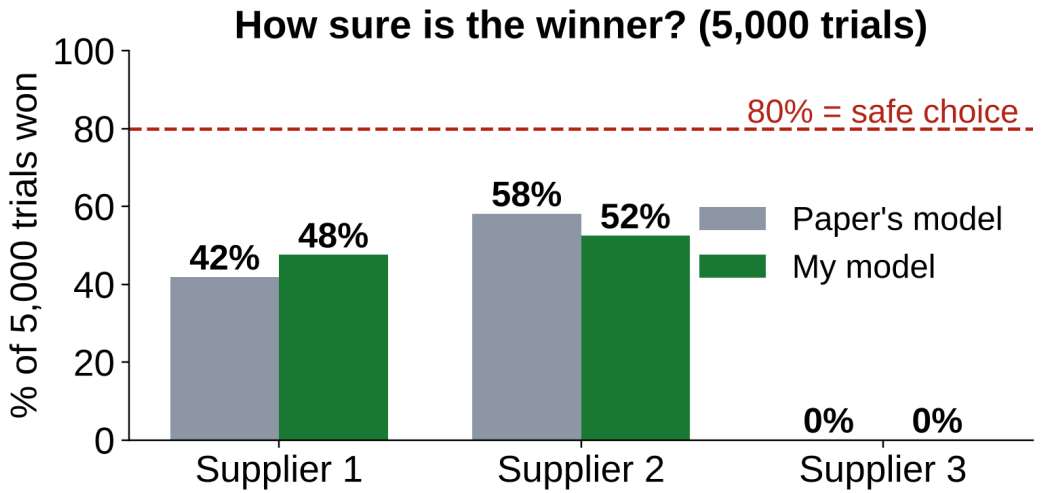
All steps are built into a simple web tool that shows the input values and the working (node server.js, then open localhost:8080).

RESULTS: PAPER'S MODEL vs MY MODEL

Paper's model



My model



	Paper's model	My model
Winner	Depends on the method. **No reason given.**	S2 0.374, S1 0.367 at w = 0.5: a **close call**, flips at w = 0.486
Consistency	**4 of 8** tables above 5%	**8 of 8** tables within 5%
Price gap	Scored as **7.3 times**	True **1.52 times** kept
How sure?	Not checked	S2 wins **52%**, S1 **48%**, S3 **0%** of trials
2-year cost	Not shown	S2 **€56.9k**, split **€72.2k**, S3 **€71.6k**, S1 **€87.5k**

Source: left, paper Tables 10 and 11; right, my web tool using the same paper Tables 2 to 9.

CONCLUSIONS AND FUTURE WORK

Conclusions

1. The AHP and ANP winners differ because of **one group weight**, not because of the network method.
2. Small, honest changes in the answers make S1 and S2 swap places. A **single ranking hides this**.
3. Fair scoring (consistent answers and real numbers) keeps **S2 just ahead**.
4. **Cost** turns the close call into a clear choice: S1 pays off only if one bad rod costs over **€5,250**.

Future work

- Fuzzy AHP for answers the managers are unsure about.
- Give managers different weights by experience.
- Use the company's real yearly data in place of estimates.
- Find the best order split by optimisation.
- Link the tool to the company's ERP data.

Recommendation: drop Supplier 3 and buy from Supplier 2. If the quality risk is high, split the order 50/50 between Supplier 1 and Supplier 2.